

Jamie Seoh

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EDUCATION

University of Waterloo

Bachelor of Software Engineering

Sep 2025 – Present

3.95/4.00 (91.18%) GPA

- Logistics Organizer @Hack the North, Prev. Sponsorship Director @Ignition Hacks & Events @UWKSA

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript & JavaScript, HTML & CSS, SQL, Java, R

Frameworks and Libraries: PyTorch, NumPy, React.js, React Native, Next.js, Svelte, FastAPI, Flask, Tailwind CSS

Technologies: Docker, GCP, Azure, AWS, Bash, MongoDB, Supabase, Git/GitHub, VS Code, Figma

EXPERIENCE

AI Software Engineer Intern

May 2026 – Aug 2026

Cerebras Systems | C++, Python, Bash, GTest

Toronto, Ontario

- Shipped C++ runtime debugging tools for inference, cutting production fault triage time by **30%**.
- Built DSL extensions to runtime config layer, enabling breakpoint injection across multi-server inference pipelines.
- Designed test infrastructure for stall detection in the inference pipeline, reaching **80% code coverage**.

Software Developer

Sep 2025 – Nov 2025

UW Orbital Satellite Mission Design Team | React, Figma, FastAPI

Waterloo, Ontario

- Built Mission Control dashboards in **React** to monitor satellite telemetry and uplink/downlink sessions.
- Implemented **REST APIs** to serve async requests to MCC, standardizing data contracts between subsystems.
- Authored unit and integration tests, achieving **95% line coverage** with **vitest** for Mission Control Center.

Front-End Developer

Sep 2023 – Apr 2024

SolveXchange | HTML, CSS, JavaScript, Flask, SQLAlchemy

Online

- Shipped and maintained responsive frontend interfaces serving **100+** active users for a non-profit platform.
- Built secure authentication and data workflows with **Flask**, **SQLAlchemy**, and **RESTful APIs**.

PROJECTS

Quota — DeltaHacks12 First Place 🏆 | TypeScript, Next.js, React Flow, Gemini, LangChain, MongoDB

- Built VS Code extension with **TypeScript** for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed codebases **<3s** for large codebases with **10,000+** lines using **AST parsing** to detect inefficiencies.
- Delivered real-time API cost insights via regex and usage caching, cutting API spend by up to **40%**.
- Created **React Flow** sandbox for budget-aware system planning with **RAG Gemini** chatbot by LangChain.

Nebula AI 🍏 | React Native, FastAPI, PostgreSQL (Supabase), Docker, GCP

- Developed a mobile app that turns handwritten notes into a searchable, **LaTeX**-formatted AI knowledge base.
- Enabled semantic search over **1,536-dim embeddings**, cutting retrieval time by **40%** with IVFFlat indexing.
- Implemented an **OCR+AI pipeline** with Gemini/OpenAI, enabling handwritten text extraction and summaries.
- Designed multi-tenant data access using Supabase Auth, JWTs, and RLS policies across 5 relational tables.
- Containerized backend with **Docker**, deployed to **Google Cloud Run** with automated CI/CD pipelines.

Personal CRM 🍏 | Next.js, FastAPI, PostgreSQL, TailwindCSS

- Built a robust full-stack CRM to centralize and prioritize personal & professional relationships at scale.
- Architected backend with **20+ REST** endpoints, enabling pagination & dynamic fetching for growing datasets.
- Improved system reliability through caching layers and rate limiting, reducing server load by **30%**.
- Designed a **matrix-based algorithm** to rank contacts, enabling users to focus on high-value relationships.

Melodie.ai 🍏 | PyTorch, NumPy, Python, Music21

- Developed an **LSTM**-based neural network with **PyTorch** to generate folk melodies, reaching coherent sequences.
- Preprocessed MIDI data by standardizing keys and normalizing sequences for simpler, faster training.
- Designed **probabilistic sampling system** with temp. control and nucleus sampling methods, enabling variation.

PUBLICATIONS

Seoh, J. (2025). *Analyzing Literary Factors: A Big Data Approach*. STEM Fellowship Journal.

Published *academic paper*: "*Literary Factors Analysis*" (DOI #: 10.17975/sfj-2025-001).